Mech 9 CW2

Smart campus project

Alaa, David, Faith, Paula, Pietro, Sanjeevan, Steven

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Project management approach

This section describes the choices we made as a group to ensure our success throughout this project in relation to our team organisation and task completion.

Overview

The primary focus was selecting a group leader to simplify roles and structure the tasks. A vote determined Steven as the leader for this project. This allowed us to set achievable goals for the team, accounting for time management and workload. Once a leader was in place, we focused on the division of tasks and our angle of approach. The group emphasised the need for improved communication, stricter deadlines, and balanced responsibilities.

Task Breakdown and Teams

The project was divided into five tasks with each task dissected for clarity and progression:

- Task 1: Stakeholders (Alaa, Sanjeevan)
  - Stakeholder research and analysis.
  - Stakeholder relevance and importance (tables/graphs) and create representative personas.
- Task 2: Straddle Vs Suspended Monorail (Paula, Pietro)
  - Research and Comparison of both systems with case studies.
  - Decision making based upon pros and cons.
- Task 3: Braking Systems (David, Faith)
  - Research multiple braking systems in the market.
  - Compare two select systems using diagrams.
- Task 4: Bridge Concept (Entire Group)
  - Use Pugh matrix to select from a range of conceptual designs.
- Task 5: Report (Steven, with group proofread)
  - Write the full project report.

Meetings and Communication Methods

Weekly meetings in EHB tracked progress, and to set deadlines for the following week. This maintained motivation for the group.

Our primary communication method were WhatsApp and OneNote, allowing us to effectively share resources and notify other group members.

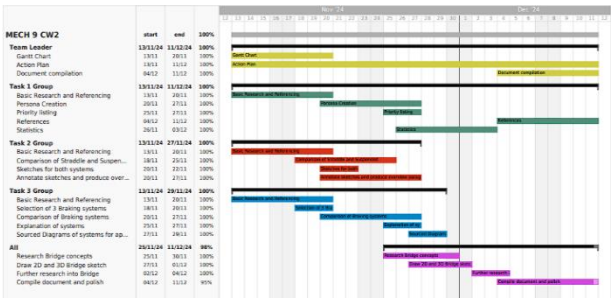
Action Plan

ID	Action	Responsibility of	Due Date
M1.1	Decide group leader	All	13/11/2024
M1.2	Create plan for project	Steven	13/11/2024
M2.1	Research Stakeholders and compile basic information	Alaa, Sanjeevan	20/11/2024
M2.2	Research Straddle Vs Suspended Monorail systems	Pietro, Paula	20/11/2024
M2.3	Research Braking systems	David, Faith	20/11/2024
M2.4	Create Gantt Chart and Action Plan, whilst writing first report pages	Steven	27/11/2024
M3.1	Go more in depth for stakeholders, creating persona and importance ranking	Alaa, Sanjeevan	27/11/2024
M3.2	Create Sketches for Suspended and Straddle Monorails with annotations	Paula	27/11/2024
M3.3	Directly Compare most common braking systems	David, Faith	27/11/2024
M3.4	Provide all references with files ready for compilation	All	27/11/2024
M3.5	Look at bridge concepts, research, and select one that you want to do	All	04/12/2024
M4.1	Start compiling Tasks 1-3	Steven	04/12/2024
M4.2	Research bridges further	All	06/12/2024
M5.1	Draw a 2D and 3D sketch of your bridge	All	06/12/2024
M5.2	Put references into the document	All	09/12/2024
M6.1	Polish Document ready for submission	All	10/12/2024

Project Organisation and Time Management

For weeks 6-9, The group was split into pairs for the first three tasks, which left the leader to organise the group project by checking on people and making a Gantt chart, Action plan for task deadlines. Weeks 9-11 were used to focus on the Bridge section collaboratively, to establish a strong foundation for future tasks. It was done by giving each person a bridge to research, allowing them to focus on a specific section, therefore permitting more comprehensive research to be done in a shorter timeframe.

Gantt Chart



Task 1: Stakeholder Analysis

Analysis of stakeholders is pivotal to the foundation of any long-term project, highlighting key considerations at every stage of development.

Persona: Student services coordinator

• Stake in monorail: Responsible for ensuring a positive student experience; transport and infrastructure are pivotal, aligning their goals with those of the University itself.

• Project Goals:

- *Increasing accessibility*: The monorail enhances campus access for those with physical disabilities or restricted mobility who would find support in a well-established system, akin to students living off-campus who benefit from shortened commute.
- *Sustainability*: The university goal of sustainability is aided by the monorail, reducing the demand for the bus service, further decreasing car emissions and optimising waste management via onboard bins.
- *Student satisfaction*: The monorail would address complaints about overcrowded buses with inconsistent, overcrowded buses characterised by long delays as an upgraded system of travel.
- *Cost-Effectiveness*: Despite the development cost, a monorail system run on electricity, including concepts such as regenerative braking is more cost-efficient.
- *Traffic*: The proposed system mitigates traffic and decreases parking space demand, futureproofing the project to align with exponential growth of students and staff.

Importance of stakeholders

A group of stakeholders was selected and placed into a comparison table (Table 1.1) to weigh up their importance at each core phase of the project. From this, the top stakeholders were further analysed, with their individual considerations stated below:

Students, Faculty and Staff:
Advocating for the project and its benefits, combined with the delivery of feedback on the project during both the Building and Operation phases allows the project to run smoothly in order with its primary users. A strong backing from students and staff encourages the financing of the project, and considering this stakeholder ensures a successful outcome. Further, the increasing student numbers establish them as a key consideration for a long-term project.[1]

Transportation Authority, Safety Regulators and Construction Companies:
More relevant to the Building and Safety and Environmental Impact phases, these stakeholders provide a key understanding of the project: funding goals, compliance with regulations and cost projections. Considering this stakeholder ensures the monorail runs safely and allows the project to be implemented in the most efficient manner.[2]

Funding bodies:
Despite ranking lower in the stakeholder analysis table, the funding bodies enable the project to run. Not only are they the primary resource for funding but play a pivotal role in ensuring the project is completed. Construction, Operation and Return on Investment are the considerations of the funding bodies, making them a crucial stakeholder to assess.[3]

"Stakeholders help maintain a clear project vision, ensuring focus on impact to achieve maximum success."

Task 2: Straddle Beam or Suspended

Methodology

This section explains our process for selecting between a suspended and straddle beam monorail. We identified the key criteria for the system, ranked their importance, and evaluated both options. The results are summarised in Table 2.1 to determine the preferred system, with sketches as supporting evidence.

Straddle-beam systems – efficient, reliable, and versatile. They allow easier evacuations whilst requiring less maintenance and are more energy efficient. They operate at higher frequencies and handle tighter turns and steeper climbs, making them easier to expand and integrate with campus infrastructure- the ideal candidate for practical and scalable projects.[4]

Suspended systems prioritise safety and environmental impact. They reduce risks for road users and use less material. Quieter systems with less ground disturbance, coupled with a

Overview

After assessing both systems, we found straddle beam to be more suited for our project, aligning with multiple key considerations of the project such as sustainability and cost-effectiveness. Furthermore, the case-studies we used matched our project goals, providing evidence of success.[6 7]

Task 3: Braking Systems

Braking Mechanisms

When braking force is applied to stop a train, it is transferred to the rails through either adhesion or non-adhesion braking. Adhesion braking uses friction between the wheels and rails, while non-adhesion methods don't rely on friction at the contact point.

Electrical and Regenerative Braking Systems

During braking, the traction motor acts as a generator, converting the train's kinetic energy into heat through a resistive circuit, acting as an electrical brake. Regenerative braking solves the issue of dissipating heat, generating reusable power from the kinetic energy instead.[8 9] Regenerative braking systems stop motion by harnessing electromagnetic induction to generate resistance in the opposite direction to the induced current, as stated by Lenz's Law

Regenerative braking systems, are economical because they do not use friction elements, unlike mechanical brake systems however The complex circuits of the electrical brake systems occasionally malfunction, so for this reason they cannot be used as emergency brakes and are ineffective at low velocities[10 11 13 18 22 25 26]

Mechanical Braking Systems

All basic mechanical braking devices use an object (a brake shoe or lining) that applies adjustable pressure and therefore friction to stop motion. Axle or wheel-mounted disc brakes are common on high-speed trains. Hydraulic Brakes fall under the umbrella of Mechanical Braking Systems[14 17 19 21]

Hydraulic Braking System

Hydraulic braking systems convert brake pedal force into pressure in an incompressible, confined fluid, creating an equal increase in pressure.[12] This actuates callipers and pistons to create friction on brake discs or drums, to slow or stopping the vehicle. By harnessing Pascal's Law or the Principle of transmission of fluid pressure, brake systems provide a safe and efficient means of deceleration resulting from their superior stopping power. [20 23 24 27]

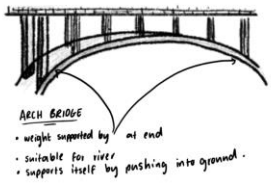
Overview

As a group we decided on regenerative braking as the best option for this project, as it proved more beneficial: Its ability to redirect power back into the motor set it ahead of the competition, alongside its many other benefits found in (Figure 3.1: Table of Comparison for Braking Systems.)

Task 4: Conceptual Design for Burleigh Brook Bridge

Our group approached this task by creating one sketch per member, ensuring each design was well-thought-out.

Concept sketch pages



ARCH BRIDGE

Advantages:

- **High load-bearing capacity:** Due to their curved shape, arches are very effective at transferring compression forces downward, making them ideal for large-span structures.
- **Aesthetics:** Arches can add a sense of grandeur and elegance to buildings and other structures.
- **Efficient water drainage:** The curved surface of an arch helps to shed water quickly, preventing it from pooling on the roof.
- **Versatility:** Arches can be constructed in a variety of shapes and sizes to suit different design requirements.

Disadvantages:

- **High construction cost:** Arch structures can be more expensive to build than other types of structures due to the complexity of their design and the specialized skills required for their construction.
- **Transportation challenges:** Large arch structures can be difficult to transport, especially by land.
- **Complex construction process:** Building an arch structure requires a high level of technical expertise and skilled labor.
- **Maintenance challenges:** Repairing and maintaining arch structures can be difficult and costly.
- **Noise concerns:** Depending on the materials used and the design, arch structures can be susceptible to noise amplification, especially during heavy rain or strong winds.

Applications:

Arch structures are used in a wide range of applications, including:

- **Industrial buildings:** Warehouses, factories, airplane hangars
- **Public buildings:** Airports, train stations, theaters, libraries, museums
- **Residential buildings:** Houses, apartments, courtyards
- **Bridges:** Arch bridges are a popular type of bridge due to their strength and aesthetic appeal.

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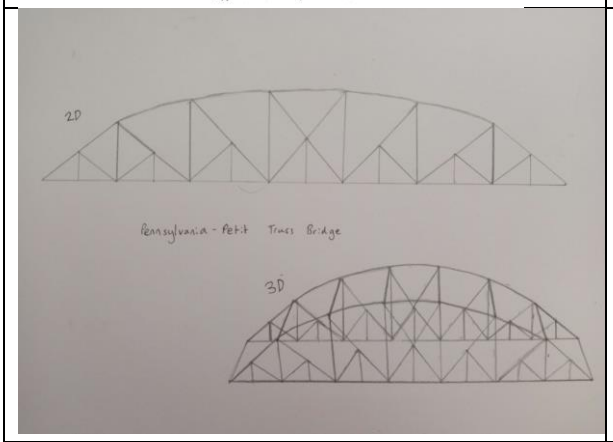

Pennsylvania Truss Bridge

Pros:

- Extremely Strong - Like most truss bridges its load bearing capacity is huge. Added support beams throughout the truss make it stronger than typically seen. Created to solve issues with truss bridges in the 19th Century where in general, design, fabrication, and erection processes became increasingly complex.
- Effective use of material - Large Truss bridges can be a very economical option.
- Can withstand extreme conditions - Large distances and varying terrain can be crossed.
- Commonly used for roadways - Versatile options for roadway implementation. It was once used for hundreds of bridges in the United States. The forms designed by Wilson and Petit in that period became the "standard" forms for US highway and railroad truss bridges, used well into the 20th century.
- Safe - Either horizontal rails or vertical pickets are the industry standard safety systems. A range of safety systems allow the user to determine the appropriate safety system for the project.

Cons:

- Requires a lot of space - Interconnecting triangles need to be large to bear and distribute heavy loads.
- High maintenance costs - Lots of parts make any truss bridge more expensive to maintain, with the Pennsylvania truss having more parts than others.
- Requires specialist construction - Can be very costly, with the complexities of the Pennsylvania Truss. This also means the bridge takes longer to implement than a simpler truss.
- They're Heavy - All truss bridges are heavy but due to the greater number of supports in the truss it can be quite heavy.
- Safety - Warren and truss bridges are the most prone to collapse.

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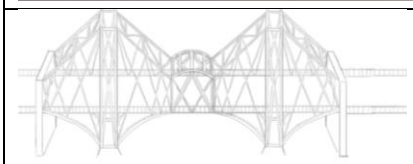

CANTILEVER BRIDGE:

Advantages:

- **Aesthetics-** Due to their long spans and truss designs they are seen as very aesthetic often used in landmarks in urban areas
- **Design-** Uses Pre-Stressed steel or concrete with no direct under supports.
- **Time to implement-** Both the multiple cantilever spans as well as the suspended bridge can both be assembled separately and attached to each other saving time

Disadvantages

- **Weight-** Requires heavier structures for stability to balance compressive and tensile forces
- **Costs-** Costs for maintenance and construction can be larger due to the size of the large structure
- **Safety-** Not suitable for earthquake-prone areas with high winds and seismic activity as well as only being suitable for small to medium vehicles
- **Environmental impact:** Due to the amount of resources needed to construct the bridge as well as the stress caused on surrounding ecosystems the overall environmental impact leans to the negative side

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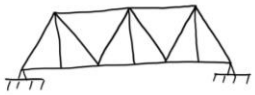

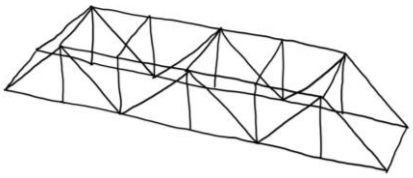

Warren Truss Bridge (With vertical members)

Advantages

1. Efficient Material Use: The Warren truss bridge design requires less material, making it cost-effective while maintaining structural integrity
2. Simple and Strong Geometry: Its triangular configuration distributes forces effectively, providing excellent stability and resistance to deformation
3. Adaptability: Adding vertical members improves the bridge's capacity to handle both distributed and variable loads
4. Ease of Construction: The simple design allows for faster and cost-effective construction

Disadvantages

1. Poor Performance Under Concentrated Loads: It does not distribute concentrated loads evenly, leading to localised stress on nearby members
2. Susceptibility to Deflection: Longer bridges may experience significant deflection, requiring additional reinforcement
3. Increased Complexity with Verticals: The addition of vertical members increases construction complexity and costs
4. Buckling Risk: Vertical members may buckle if not designed with sufficient stiffness

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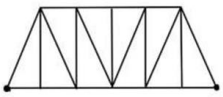

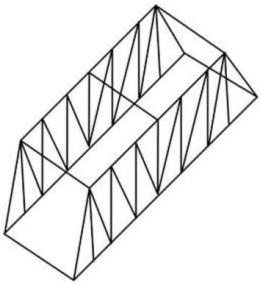

Pratt Truss Bridge

Advantages

1. The design addresses all member's behaviour – diagonal members are in tension, vertical members in compression
2. Pratt Truss Bridge design is a cost-effective structure that is easier to build by unskilled labour, it has a low initial cost and is easy to transport.
3. The design of Pratt Truss Bridge can be applied in different design situations – the design can be modified to reduce the span-to-depth ratio.
4. The design of Pratt Trus Bridge has several built examples, and efficient manufacturing methods for its construction have been developed through various iterations.
5. Simple design – fewer members.
6. Structural behaviour is well understood – members are simply in tension and compression; therefore, the behaviour can be analysed using simple analysis tools.
7. The bridge is straight and smooth – ideal for heavy traffic such as railroads, trucking and modern cars.
8. Utilizes standard parts, which means the design can be quickly assembled in a span-assembly shop to reduce field costs.
9. Well accepted and used design and is widely used in bridge building world.

Pratt Truss Bridge Disadvantages

1. Pratt Truss Bridge is not as advantageous if the load is not vertical (but diagonal); vertical and horizontal loads require different design considerations.
2. If not well designed, Pratt Truss Bridge can be susceptible to failure– which can result in high maintenance costs.
3. Not efficient in long span and shallow depth situations – in these situations other truss designs are preferred.
4. Aesthetically Pratt Truss Bridge is not as pleasing as other truss designs – although this can depend on personal taste.

Cost

The Pratt truss is known for its cost-effectiveness due to efficient material usage. It employs diagonal and vertical beams arranged in a triangular pattern, minimizing the amount of material required without compromising strength. This reduces construction costs compared to other designs

Design

The Pratt truss uses diagonal members that experience tension and vertical members that handle compression. This configuration is especially effective for uniformly distributed loads. It can span medium-to-large distances and is suitable for both roadways and railways

Safety

The structural design of the Pratt truss ensures even distribution of loads, enhancing stability. The tension in diagonal members reduces stress concentrations, contributing to overall safety and durability, even under heavy loads like trains or trucks

Weight

Using materials efficiently makes the Pratt truss lighter than some alternative designs. Modern

Effectiveness

The bridge is highly effective for spans up to 200 feet, balancing simplicity, strength, and versatility. It has been a standard for many infrastructure projects worldwide

Time to Implement

The straightforward design speeds up construction compared to more complex structures. Prefabrication of truss components can further reduce build time

Environmental Impact

The environmental impact depends on the materials used. Traditional designs using wood or modern steel variants have moderate impacts. However, efficient material use in the Pratt truss lowers its carbon footprint compared to heavier designs

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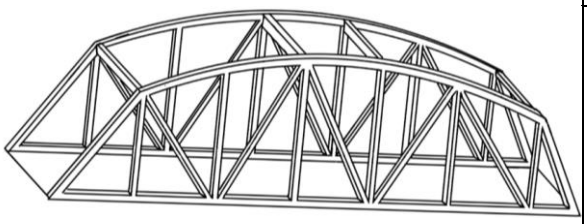

Camelback Warren Truss With Verticals

Advantages

- Triangles spread load evenly between members making them very suitable for distributed loads
- Easier to construct than others like the Pratt or Howe trusses
- They are lighter and have less wind pressure due to the air gaps between members
- Due to the arch bridges can span a greater distance
- Multiple arch's can work together to increase stability

Disadvantages

- More challenging to design and analyse
- Is inefficient with concentrated point loads as the forces are concentrated
- Arch makes it unsuitable for some locations
- Arch is expensive to construct

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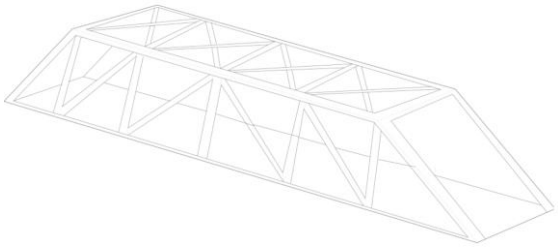

HOWE TRUSS BRIDGE

Benefits

1. **Strength/Load-Bearing Capacity**: The triangular design effectively distributes weight and balances compression and tension, making it suitable for heavy loads such as railroads or highways
2. **Versatility and Durability**: The Howe truss can span large distances and withstand extreme conditions, including heavy traffic and adverse weather, making it adaptable for diverse environments
3. **Simplicity of design**:
   While the diagonal supports are in compression, the vertical supports are in tension, simplifying the analysis and design process. The design is modular, meaning it can be scaled to various spans and customized for different applications.

Drawbacks

1. **Large Space Requirement**:
   Its design requires ample room, which may not be ideal for constrained areas
2. **Weight**:
   The significant use of materials makes it heavy, which might require additional support structures and increase costs in areas with weaker foundations
3. **Complex Construction and Maintenance**:
   The intricate design may lead to higher construction and maintenance complexity compared to simpler bridge designs. This might also reduce its aesthetic appeal.
4. **Susceptibility to Vibration**:
   The rigidity of the structure can lead to vibrations under dynamic loads, which may require additional damping measures for stability

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Bridge Selection

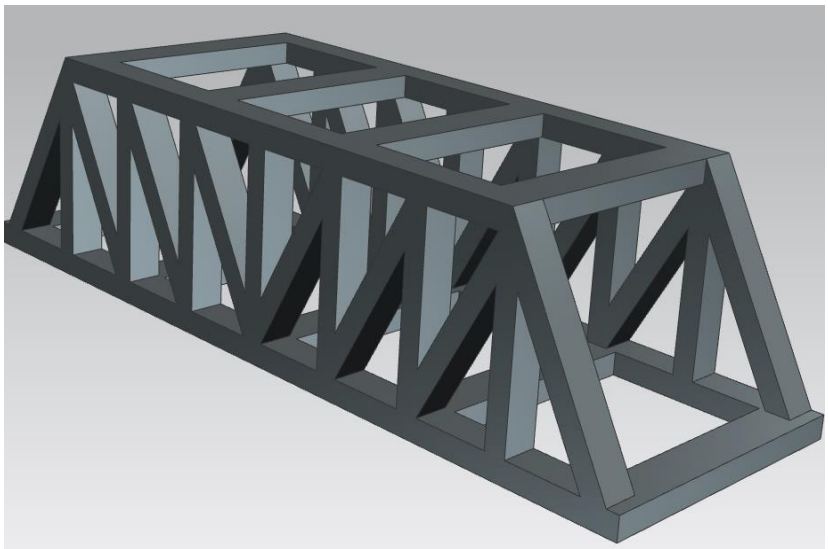
After conducting in-depth analysis and constructing a Pugh matrix with the Pennsylvania Truss as the 'Current Solution', we determined that the Pratt Truss Bridge is the optimal choice for the Burleigh Brook problem. Its strength-to-weight ratio, simplicity, low environmental impact, and quick implementation align with both project and university goals.

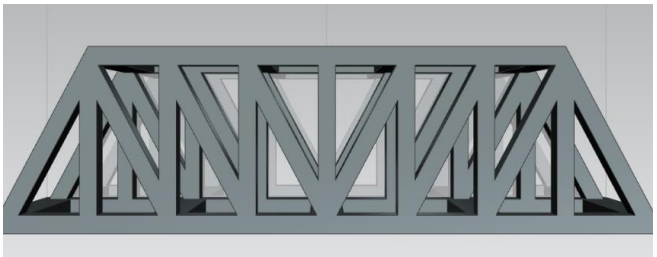
Criteria	Weighting	Current Solution	Arch Bridge	Cantilever Bridge	Warren Truss with Verticals	Pratt Truss	Camelback Warren Truss	Howe Truss
Cost to implement	2	0	-	-	+	+	0	+
Design	1	0	+	+	0	0	+	0
Safety	5	0	+	-	+	+	+	-
Weight	3	0	-	-	+	+	0	+
Effectiveness	4	0	+	0	-	+	0	+
Time to implement	2	0	+	-	+	+	+	+
Environmental impact	3	0	+	-	+	+	0	0
Total	N/A	0	10	-14	11	19	8	6

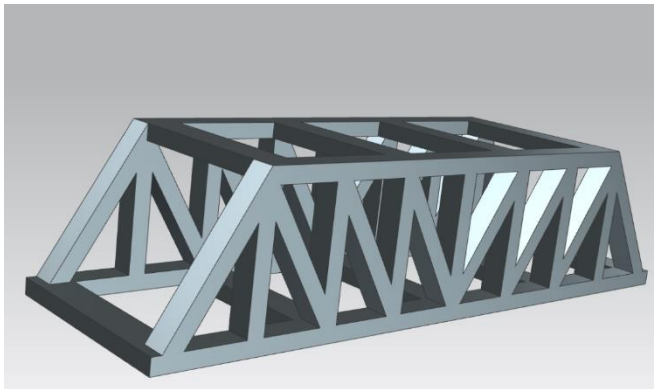
We opted for a Weighting of Importance from 1-5 for the criteria, to ensure that the comparison between bridges was fair, and matched the key focuses of the project (Safety being most important, and Design).

3D Model of Selected Bridge

The images below showcase a 3 d rendering of the selected bridge design, aiding in the visualisation







Appendices

Figure 1.1: Table of Stakeholders Ranked by importance

Stakeholders:	Student Services	Students, Faculty and Staff	Sporting Teams	Local Transportation Authority	Safety Regulators	Construction Companies	Funding Bodies
Financing	2	3	4	5	7	6	1
Building	6	5	7	2	3	1	4
Safety and Environmental impact	6	3	7	2	1	4	5
Operation	3	1	2	4	5	7	6

Figure 1.2: Statistical Analysis of University Student data

Academic year	Full time students	Full time non-international students	International stud	Part time student	Total number of students	Year on year growth
2014-15	13,732	11,395	2,337	1,674	15,606	
2015-16	14,922	12,019	2,903	1,866	16,788	8
2016-17	15,374	12,472	2,902	1,801	17,175	2
2017-18	15,634	12,498	3,136	1,503	17,137	- 0
2018-19	16,563	13,182	3,381	1,327	17,890	4
2019-20	17,077	13,436	3,641	1,228	18,305	2
2020-21	17,097	14,227	2,870	1,205	18,302	- 0
2021-22	17,839	14,486	3,353	1,320	19,159	5
2022-23	18,160	14,488	3,672	1,303	19,463	2
2023-24	17,882	14,372	3,510	1,166	19,048	- 2

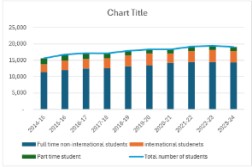
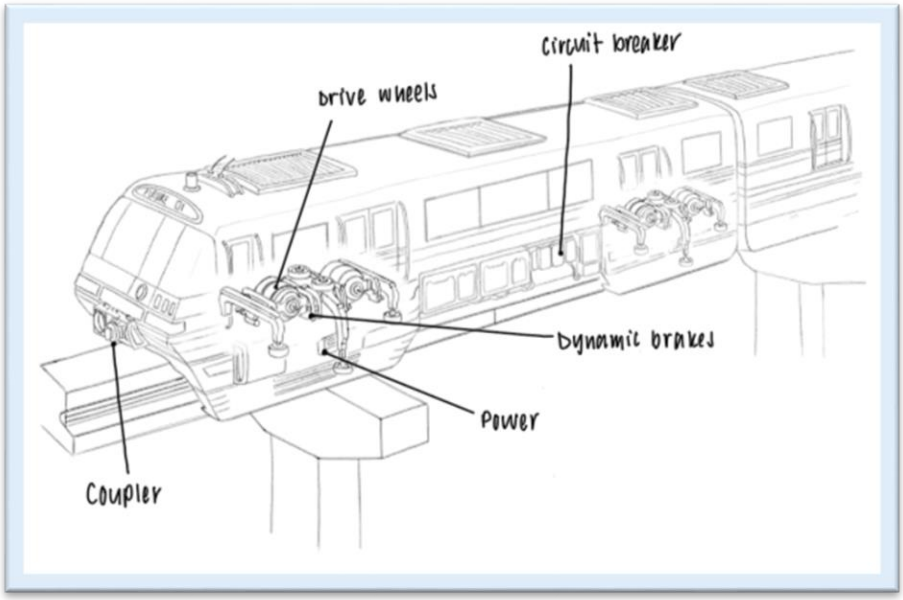


Figure 2.1: Table of comparison Straddle vs Suspended Beam

Table 1.1		Straddle Beam	Suspended	Details
Safety		**=**	**=**	
	Evacuation Procedures	+	-	Straddle Beam allows easier on-track evacuation compared to Suspended systems.
	Weather Resistance	+	-	Straddle Beam systems generally offer better wind resistance.
	Natural Disaster Safety	-	+	Suspended systems are less prone to derailment during earthquakes.
	Safety of surrounding	-	+	Suspended systems are high up in the air, posing less risk for other road users
Cost		**+**	**+**	
	Building Cost	+	-	Straddle beam systems are built with less materials and more easily
	Operational Cost	=	=	Same for both systems
	Maintenance Cost	+	-	Straddle beam systems require less maintenance and offer an easier maintenance procedure
Sustainability		**=**	**=**	
	Building CO2 releases	-	+	Suspended systems use less material, reducing construction emissions.
	Operational CO2 releases	+	-	Straddle Beam systems are more energy-efficient per passenger.
	Intrusion for surrounding vegetation	-	+	Suspended systems have less ground-level disturbance.
	Energy efficiency	+	-	Straddle Beam systems have better efficiency.

Performance		**+**	**-**	
	Passengers per Car	+	-	Straddle Beam systems typically have larger passenger capacities per car.
	Frequency	+	-	Straddle Beam systems can operate at higher frequencies due to stability and lower internal beam stress.
	Monorail Speed	+	-	Straddle Beam systems achieve higher speeds easier build.
	Noise	-	+	Suspended systems tend to generate less noise during operation.
	Turning radius	+	-	Straddle beam systems have a tighter turning radius
	Height Climb Rate	+	-	Straddle beam systems have a higher climbing rate
Generals		**+**	**-**	
	Accessibility	+	-	Straddle Beam systems have more convenient boarding platforms.
	Space efficiency	-	+	Suspended systems use less space and have thinner pillars.
	Scalability	+	-	Straddle Beam systems are easier to expand due to modular track designs.
	Design	-	+	Suspended systems are more visually appealing
	Campus Integration	+	-	Straddle Beam systems Integrate better with campus infrastructure
	Route Implementation flexibility	O	O	Both are challenging to implement on campus
Overall		**+**	**-**	Straddle beam system is more suited for our project

Figure 2.2 and 2.3: Sketch of Straddle Beam Monorail and Sketch of Suspended Monorail



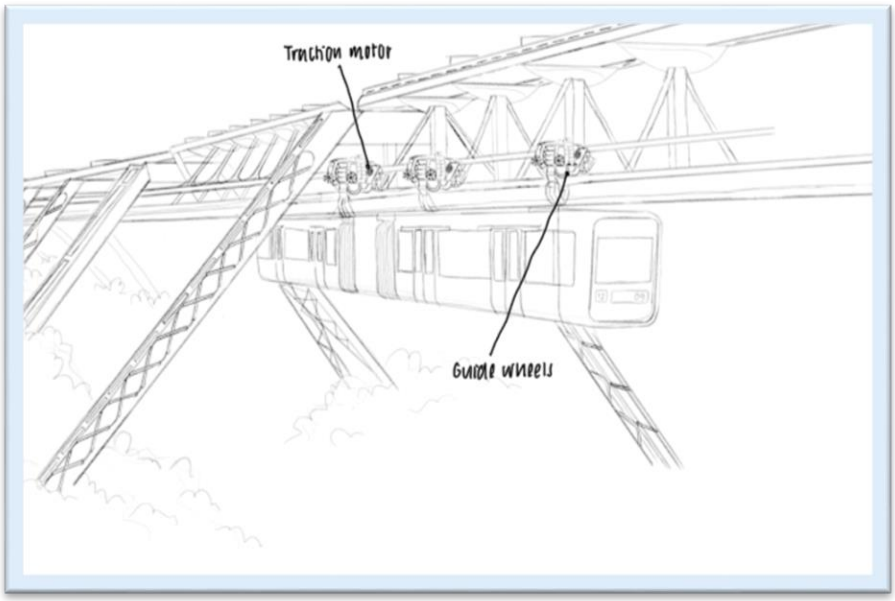


Figure 3.1: Table of Comparison for Braking Systems

	ELECTROMAGNETIC	REGENERATIVE	HYDRAULIC
PROS	• Rapid response times allowing for quick engagement and disengagement • Highly reliable and consistent • Enhanced safety due to lack of hydraulic lines	• Electricity generated from kinetic energy is retransmitted • More environmentally friendly	• Simple construction and maintenance • Low break wear rate • Equal and consistent braking action on all wheels
CONS	• Limited heat dissipation • Lower torque capacities	• Less effective at low speeds • Occasional malfunctions due to complex circuitry	• Fluid leakage could cause system failure • Requires additional maintenance • Air contamination affects brake efficiency

Figure 3.2: Regenerative Braking System In-depth Sketch

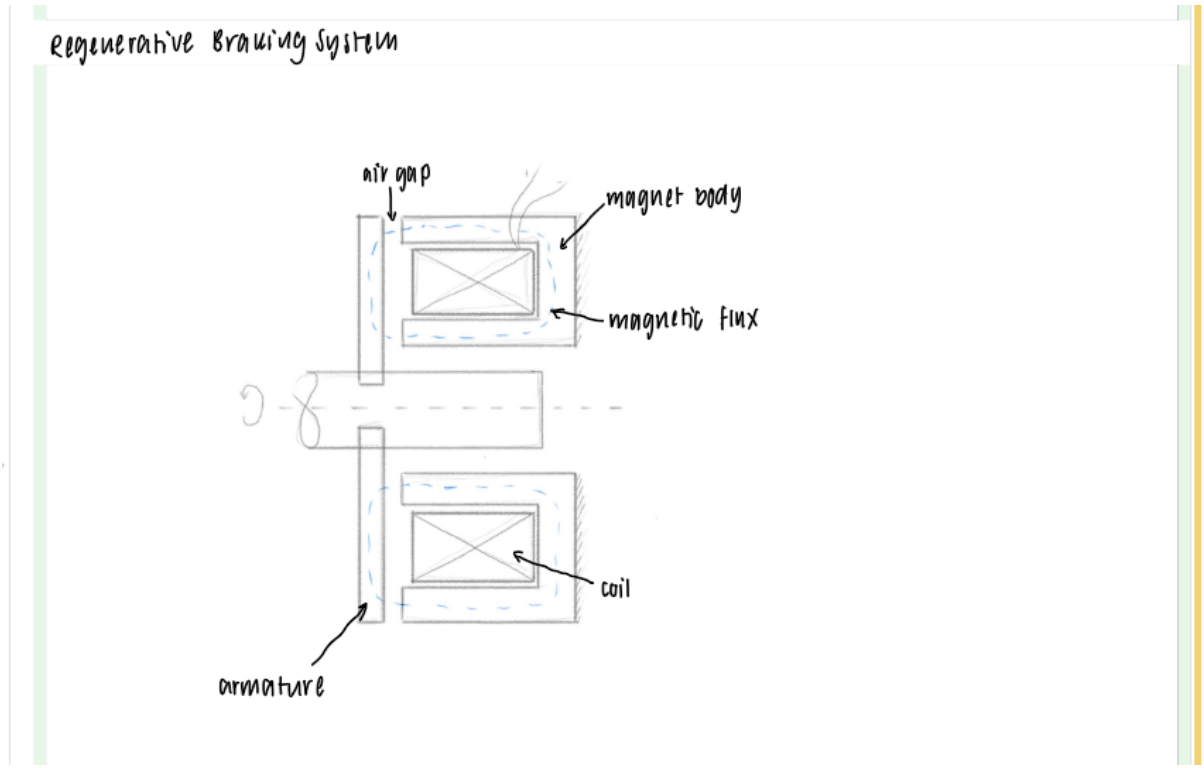


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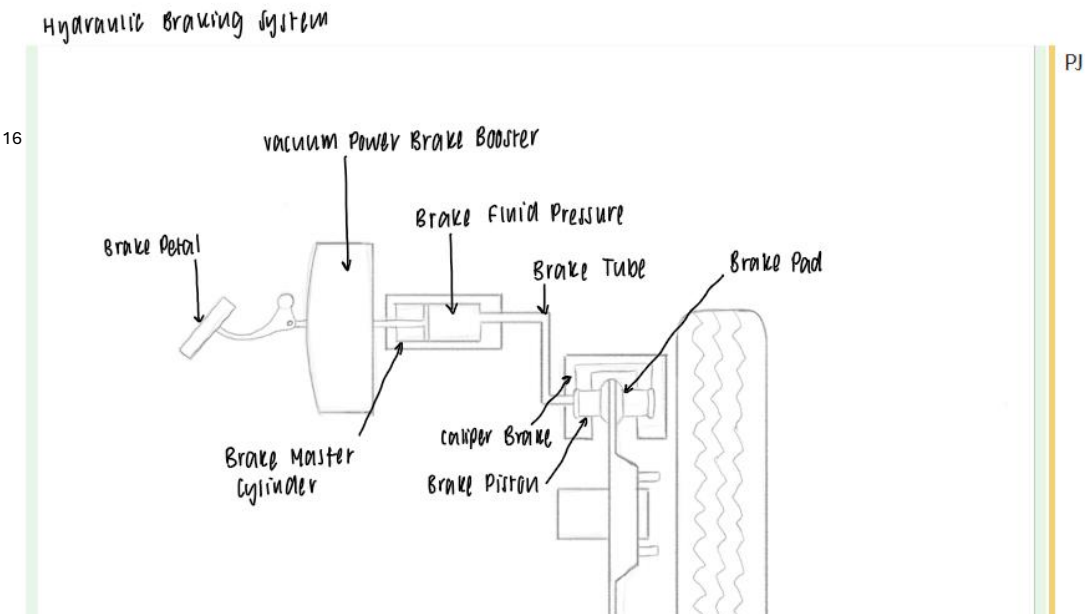
Figure 3.3: Hydraulic Braking System Sketch



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